

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 14

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. The Producer-Consumer code discussed in Lecture 14 utilizes a `while(1)` loop, representing an infinite workload. Rewrite the implementation to handle a finite workload of N items and modify the logic to ensure that both the Producer and Consumer threads terminate cleanly once all work is completed.
2. Suppose you have 5 producer threads and 8 consumer threads competing for the same 10-slot buffer.
 - A. Why is it critical that the signal occurs while the state is protected by the mutex in a multi-threaded pool?
 - B. A thread might wake up and find the buffer empty again because another woken thread stole the item first. Explain how the `while` loop protects against this specific race condition.
 - C. If you have 3 producers waiting for space, which condition variable should the consumer signal after extracting an item: `empty` or `full`?
 - D. Rewrite the producer-consumer logic to support a system with multiple producer threads and multiple consumer threads sharing a single bounded buffer of size 10.

- **Welcome: Sean**

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